ISA/IEC 62443-3-3 / 4-2 • SL-3 TESTING CHEAT SHEET

ProSoft MVI56E-MNET / MNETXT Modbus TCP/IP Module (in-chassis, ControlLogix) · Component type: **Embedded Device (EDR) - in-chassis Modbus TCP/IP communication module** · Target SL = SL 3

Where to work	In-chassis 1756 ControlLogix module **owned by the ControlLogix processor** (data exchanged over the backplane via a **Studio 5000 Add-On Instruction (AOI)** + controller tags). Configure with **ProSoft Configuration Builder (PCB)** over the module's **Ethernet port** (or via **CIPconnect** through a 1756-EN2T/EN4TR); **ProSoft Discovery Service (PDS)** assigns a temp IP; an on-board **web server** serves module docs/sample. Modbus is **Modbus TCP/IP**: client (master) + server (slave) on **port 502 (MBAP)**, up to 10 server connections. NOTE: **the same Ethernet port carries Modbus TCP AND config/diagnostics** - segment tightly. Harden = restrict Modbus clients, disable the web, and let the ControlLogix + **FactoryTalk Security** govern the processor side.
Ports / paths	**Modbus TCP/IP** on the Ethernet port: **TCP 502 (MBAP)** - client + server (server = up to 10 connections). The **same Ethernet port** also serves config/diagnostics: **PDS discovery**, **PCB config**, on-board **HTTP web**, **CIPconnect** bridging. **1756 backplane** (AOI data exchange). 4-char LED status display. No TLS/crypto (unauthenticated Modbus TCP + HTTP).
Notes / lifecycle	In-chassis MVI56E-MNET / MNETXT (XT = extended temperature). No user accounts, no comms encryption. Native controls = **restrict which Modbus TCP clients can reach 502**, **disable the on-board web**, protect the PCB **.ppf**, the owning **ControlLogix + FactoryTalk Security / keyswitch**, and physical control. Because Modbus TCP and config share one Ethernet port, **segmentation is the primary control**. Confirm firmware; track ProSoft / CISA advisories.
How to read	Default read-only (PCB read-only config review; read-only Modbus TCP reads; module LED/web diagnostics; Studio 5000 AOI/tag review). Each row = native step-by-step + command-level external-tool validation + pass. Write/exploit tools = isolated bench + written authorization. Several rows are Compensating / N-A-at-device.

FR 1 — Identification & Authentication Control (IAC)

Req (3-3 / 4-2)	Requirement	Where & how to verify (native steps + Ext. tools)	Pass criteria / evidence	Notes / lifecycle
SR/CR 1.1	Human user identification & auth	The module has **no user accounts**. Two access surfaces on the **one Ethernet port**: (1) config - a host that reaches the port can reconfigure via **PCB** (PDS + the on-board web have minimal/no auth); (2) operational **Modbus TCP 502** - any reachable client can read/write the server. Compensate by **segmenting the port** and restricting Modbus clients; the **processor side** (backplane data) is governed by the **ControlLogix + FactoryTalk Security**. **Test:** from an unauthorized host, open Modbus TCP 502 / run PDS-PCB -> blocked by segmentation; browse the module web -> docs only. **Ext. tools:** `modpoll -m tcp -a` from an unauthorized segment (expect blocked); `curl -i http:///`; FactoryTalk Security (processor-side).	Ethernet port segmented + Modbus clients restricted; processor-side FactoryTalk; no anonymous reconfig/write.	EDR - no module accounts; compensating.
SR/CR 1.2	Device & software-process I&A;	Modbus TCP is **unauthenticated**; the module is identified by its **IP** (and Modbus unit ID as server). Bind identity by restricting which **client IPs** may open 502 (firewall / the module's allowed-client scope) + segmentation; document the authorized master(s). **Test:** from a non-authorized client IP, open a 502 connection -> blocked (firewall/segmentation). **Ext. tools:** `modpoll -m tcp -a` (read-only, authorized); `nmap -p 502` ; confirm IP + unit ID.	Module identified by IP/unit ID; client access restricted; unauthenticated protocol compensated.	No device auth - Compensating.
SR/CR 1.3	Account management	No module accounts; manage segment access to the Ethernet port, the PCB **.ppf** custody, and **FactoryTalk Security** on the owning processor (least privilege, no shared logins). **Ext. tools:** Segment/firewall access review; .ppf custody record; FactoryTalk Security (user/role).	Port access + .ppf + processor identities controlled; defaults removed.	
SR/CR 1.4	Identifier management	Confirm a unique module **IP** + Modbus unit ID, and a unique module identity (slot + serial); no reuse on the segment. **Ext. tools:** PCB (IP/unit ID); `nmap -p 502` for collisions; Studio 5000 (module identity); asset register.	Unique IP/unit ID + module identity; no reuse.	
SR/CR 1.5	Authenticator management	The module stores **no credentials** and the config/web have no strong auth - compensate with segmentation + client restriction; enforce **FactoryTalk Security** password policy on the owning processor. **Ext. tools:** Confirm segmentation + client restriction; FactoryTalk password policy.	No module credentials (segmentation compensates); processor policy enforced.	Segmentation-based.

Req (3-3 / 4-2)	Requirement	Where & how to verify (native steps + Ext. tools)	Pass criteria / evidence	Notes / lifecycle
SR 1.6	Wireless access management	**N/A.** Wired in-chassis module with no wireless interface. **Ext. tools:** Confirm no wireless.	N/A - justification recorded (wired device).	N/A
SR/CR 1.7	Strength of password-based authentication	Confirm the password-strength policy (length/complexity/lifetime) meets SL-3. **Steps:** for this product this is assessed with the same evidence as **SR/CR 1.5** above (shared steps); confirm this requirement individually and adjudicate it separately.	Strength of password-based authentication verified individually.	
SR/CR 1.8	Public key infrastructure (PKI) certificates	**N/A at device.** No PKI/TLS/crypto (Modbus TCP + HTTP). Record N/A; integrity/confidentiality by segmentation. **Ext. tools:** Confirm no certificate/crypto; Wireshark (Modbus TCP + HTTP cleartext).	N/A device PKI; compensating segmentation.	N/A.
SR/CR 1.9	Strength of public key-based authentication	Confirm public-key authentication uses strong, current keys/algorithms. **Steps:** for this product this is assessed with the same evidence as **SR/CR 1.8** above (shared steps); confirm this requirement individually and adjudicate it separately.	Strength of public key-based authentication verified individually.	
SR/CR 1.10	Authenticator feedback	**N/A at device** (no module login). Applies to FactoryTalk on the processor. **Ext. tools:** FactoryTalk visual verification.	N/A at device; FactoryTalk masks entry.	N/A.
SR/CR 1.11	Unsuccessful login attempts	**N/A at device** (no login). Remove exposure with segmentation + client restriction; FactoryTalk lockout on the processor; monitor anomalous Modbus TCP. **Ext. tools:** FactoryTalk lockout; Modbus/OT IDS.	No device login; segmentation + processor lockout + monitoring.	Compensating.
SR/CR 1.12	System use notification	**N/A at device.** Placard on the cabinet; banner on the Studio 5000 EWS (compensating). **Ext. tools:** Physical placard; EWS banner.	Use notice via placard/EWS (compensating).	Compensating.
SR 1.13	Access via untrusted networks	The Ethernet port carries **Modbus TCP 502 + config** - it must be on a controlled OT segment, never untrusted/routable. Firewall 502 to authorized client IPs only, block config (PDS/web) from operational access, and reach config via a controlled path/CIPconnect; no untrusted reach. **Test:** from an untrusted VLAN, reach 502 / the web -> must fail. **Ext. tools:** `nmap -Pn -p 502,80` from the untrusted side (expect filtered); firewall/allowed-client review.	Module unreachable from untrusted networks; 502 restricted to authorized clients; config path controlled.	

FR 2 — Use Control (UC)

Req (3-3 / 4-2)	Requirement	Where & how to verify (native steps + Ext. tools)	Pass criteria / evidence	Notes / lifecycle
SR/CR 2.1	Authorization enforcement	No module roles. Authorization is (a) **restrict which client IPs may write via Modbus TCP 502** and scope the writable server registers in the PCB map, and (b) the **ControlLogix processor + FactoryTalk Security / keyswitch** govern the backplane data (AOI/tags) and config. **Test:** from an unauthorized client, attempt a Modbus write -> blocked; confirm read-only server points cannot be written. **Ext. tools:** PCB (server write-scope / allowed clients); `modpoll` write test (authorized/isolated); FactoryTalk (processor authz).	Writes restricted by client + scoped registers; processor governed by FactoryTalk.	No device roles - Compensating.
SR/CR 2.2	Wireless Use Control	**N/A.** No wireless interface on this device; wireless access is managed at the network layer.	N/A - justification recorded (not assessed as a control).	N/A
SR/CR 2.3	Use control for portable/mobile devices	Control the PCB laptop + the **.ppf** config file; only authorized firmware loaded; the module is in a chassis - control physical access. **Ext. tools:** Host/media control; .ppf ACL; physical inspection (chassis).	Engineering media + config file controlled.	
SR/EDR 2.4	Mobile code	**N/A.** Fixed-function module firmware executing the compiled config - no general-purpose or downloadable code. **Ext. tools:** Confirm fixed-function device (no code path beyond firmware).	N/A - no mobile-code capability (fixed firmware).	EDR-specific / N/A.
SR/CR 2.5	Session lock	**N/A at device** (no interactive session). EWS OS lock for PCB. **Ext. tools:** EWS inactivity-lock GPO.	N/A at device; EWS locks.	N/A.

Req (3-3 / 4-2)	Requirement	Where & how to verify (native steps + Ext. tools)	Pass criteria / evidence	Notes / lifecycle
SR/CR 2.6	Remote session termination	The server bounds concurrency at **10 connections** - confirm that is sufficient and monitored, that idle connections close, and that PCB diagnostic sessions do not conflict with operational polling. **Ext. tools:** PCB (connection/diagnostics); `nmap -p 502`; Wireshark (connection count / teardown).	Modbus TCP concurrency bounded (10) + monitored; sessions terminable.	
SR/CR 2.7	Concurrent session control	Confirm concurrent sessions are bounded/limited and monitored. **Steps:** for this product this is assessed with the same evidence as **SR/CR 2.6** above (shared steps); confirm this requirement individually and adjudicate it separately.	Concurrent session control verified individually.	
SR/CR 2.8	Auditable events	The module has **no native security log**. Capture change attribution at the **ControlLogix processor + FactoryTalk AssetCentre**, the PCB **.ppf version**, and a Modbus-TCP-aware **OT IDS** (connection/write events on 502); forward to the SIEM. **Ext. tools:** AssetCentre (processor/module change); PCB .ppf version control; Modbus/OT IDS + SIEM.	Changes attributable at processor/AssetCentre + .ppf version; Modbus TCP events captured by IDS.	No native log - Compensating.
SR/CR 2.9	Audit storage capacity	Logging is off-module; confirm the processor/AssetCentre/SIEM are sized/retained and alert on collection or comms-loss failure. **Ext. tools:** AssetCentre/SIEM retention + comms-loss alert.	Off-module logs sized/retained; audit-failure/comms-loss alerting defined.	Compensating.
SR/CR 2.10	Response to audit processing failures	Confirm a defined response to audit-processing/forwarding failures (alerting; no silent loss). **Steps:** for this product this is assessed with the same evidence as **SR/CR 2.9** above (shared steps); confirm this requirement individually and adjudicate it separately.	Response to audit processing failures verified individually.	
SR/CR 2.11	Timestamps	The module has no RTC; events are timestamped by the ControlLogix processor / SCADA - confirm those are time-synced (CIP time / NTP) so module events correlate. **Ext. tools:** Processor WallClockTime / SCADA NTP check.	Correlatable time via the synced processor/SCADA.	
SR/CR 2.12	Non-repudiation	Confirm actions are attributable to a specific user (non-repudiation). **Steps:** for this product this is assessed with the same evidence as **SR/CR 2.8** above (shared steps); confirm this requirement individually and adjudicate it separately.	Non-repudiation verified individually.	
EDR 2.13	Physical diagnostic & test interfaces	Control physical interfaces: the module in a **locked chassis/cabinet**; the **Ethernet port** (segment tightly), and the backplane; secure the network drop. **Ext. tools:** Physical inspection (chassis lock, Ethernet drop); confirm segment/firewall.	Physical/diagnostic interfaces controlled; Ethernet port tightly segmented.	EDR-specific.

FR 3 — System Integrity (SI)

Req (3-3 / 4-2)	Requirement	Where & how to verify (native steps + Ext. tools)	Pass criteria / evidence	Notes / lifecycle
SR/CR 3.1	Communication integrity	Modbus TCP (502) and the on-board web are **unauthenticated cleartext** (no TLS) - integrity/confidentiality come from **containment**: keep the module on a segmented OT cell, restrict 502 to authorized client IPs, and **disable the on-board web**; no cleartext control/config crosses a zone boundary uncontrolled. **Test:** confirm Modbus TCP stays in-zone to authorized clients; the web is off/segmented. **Ext. tools:** Wireshark (`modbus` cleartext in-zone; HTTP); `nmap -p 502,80` (502 to authorized only; web off).	Modbus TCP contained/segmented to authorized clients; web disabled; no uncontrolled cleartext.	No crypto - Compensating.
SR/EDR 3.2	Protection from malicious code (component)	Confirm only ProSoft firmware runs (fixed-function). Load firmware only from ProSoft via PCB and verify provenance; restrict who can update (segmentation); track ProSoft / CISA advisories. **Ext. tools:** PCB (firmware version + provenance); ProSoft advisory review.	Only vendor firmware runs; provenance verified; update restricted.	EDR-specific.
SR/CR 3.3	Security functionality verification	After any change re-verify: 502 client restriction, server write-scope, web disabled, IP config, firmware, and processor-side FactoryTalk, against the baseline; compare the PCB **.ppf**. **Ext. tools:** PCB (config vs baseline .ppf); `nmap -p 502,80` ; FactoryTalk review.	Security settings match the baseline after change.	

Req (3-3 / 4-2)	Requirement	Where & how to verify (native steps + Ext. tools)	Pass criteria / evidence	Notes / lifecycle
SR/CR 3.4	Software & information integrity	Protect the PCB **.ppf** config and the **AOI/controller tags** in the Studio 5000 project: host ACLs, version control; hash the .ppf; the ControlLogix keyswitch/FactoryTalk prevent processor-side change; detect drift by compare. **Ext. tools:** Host ACL; `certutil -hashfile .ppf SHA256`; Studio 5000 online-vs-offline compare.	Config + AOI integrity protected; unauthorized change prevented/detected.	
SR/CR 3.5	Input validation	Confirm the module tolerates malformed **Modbus TCP** frames without faulting or corrupting the map. **ISOLATED bench only:** fuzz 502 on a bench module and watch the LED/AOI status. **Ext. tools:** Modbus-TCP fuzzer (`boofuzz`) on a bench module; watch LED status + AOI status word.	No fault or map corruption on malformed input.	Do NOT fuzz an in-service module.
SR/CR 3.6	Deterministic output	Verify **comm-fail behavior**: on Modbus TCP loss the module signals status to the AOI, and the **ControlLogix logic** drives the process to a defined state (a stale Modbus read must not present as live). Confirm the server/client timeout + comm-fail handling. **Ext. tools:** PCB (comm-fail/timeout config); Studio 5000 (AOI status -> fault logic); simulate 502 loss.	Comm loss signaled to the processor; process assumes a defined state (no stale data as live).	
SR/CR 3.7	Error handling	Confirm graceful error handling: the module **LED display** + the AOI status/error words report faults without unsafe behavior or info leak. **Ext. tools:** Module LED review; Studio 5000 (AOI status/error words).	Graceful error handling; no unsafe state or info leak.	
SR/CR 3.8	Session integrity	**N/A / limited.** No authenticated/encrypted Modbus TCP session; integrity by containment. Record N/A + compensating segmentation. **Ext. tools:** Confirm no session security; segmentation review.	N/A/limited at device; compensating segmentation recorded.	N/A/Partial.
SR/CR 3.9	Protection of audit information	Since audit is off-module, protect the AssetCentre / IDS / SIEM store and restrict who can read/clear it. **Ext. tools:** AssetCentre / SIEM access review.	Off-module audit records protected from unauthorized access.	
EDR 3.10	Support for updates	Update firmware via **PCB** with firmware from ProSoft; verify version/provenance; plan in a maintenance window with a config (.ppf) restore/rollback plan. **Ext. tools:** PCB (update flow + provenance); ProSoft firmware; rollback/.ppf restore plan.	Controlled updates from a verified source, with rollback.	EDR-specific.
EDR 3.11	Physical tamper resistance/detection	Keep the chassis in a locked/sealed cabinet; control the Ethernet drop and the module; config drift is visible via PCB compare / AssetCentre. **Ext. tools:** Physical inspection (cabinet/seal, Ethernet drop); PCB/AssetCentre compare.	Chassis secured; interfaces controlled; drift detectable.	EDR-specific.
EDR 3.12	Provisioning product-supplier roots of trust	**N/A.** No documented signed-firmware supplier root of trust. Record N/A; compensate with firmware provenance (3.10), segmentation, physical control. **Ext. tools:** Confirm no RoT; ProSoft provenance process.	No supplier RoT; compensating controls recorded.	N/A.
EDR 3.13	Provisioning asset-owner roots of trust	**N/A.** No asset-owner certificate provisioning (no PKI). Record N/A; asset-owner trust at the network + physical + processor layer. **Ext. tools:** Confirm no cert provisioning; segmentation review.	No asset-owner RoT; compensating controls recorded.	N/A.
EDR 3.14	Integrity of the boot process	**N/A.** No verified/secure boot. Record N/A; mitigate via firmware provenance (3.10), segmentation, physical control, and monitoring. **Ext. tools:** Confirm no secure-boot capability.	No verified boot; compensating controls recorded.	N/A.

FR 4 — Data Confidentiality (DC)

Req (3-3 / 4-2)	Requirement	Where & how to verify (native steps + Ext. tools)	Pass criteria / evidence	Notes / lifecycle
SR/CR 4.1	Information confidentiality	Confirm information confidentiality — sensitive data is protected in transit and at rest. **Steps:** for this product this is assessed with the same evidence as **SR/CR 3.1** above (shared steps); confirm this requirement individually and adjudicate it separately.	Information confidentiality verified individually.	
SR/CR 4.2	Information persistence	On decommission/repurpose, clear the module config (download a default/blank .ppf) and destroy/secure the **.ppf** + backups so the IP, Modbus maps, and command lists are not recoverable. **Ext. tools:** PCB (download default/blank); .ppf + backup sanitize.	Module config + maps purged on decommission or reuse.	

Req (3-3 / 4-2)	Requirement	Where & how to verify (native steps + Ext. tools)	Pass criteria / evidence	Notes / lifecycle
SR/CR 4.3	Use of cryptography	**N/A.** No comms crypto on the module. Rely on segmentation; where confidentiality is required, add an external secure layer. **Ext. tools:** Confirm no crypto; segmentation review.	No comms crypto - compensating segmentation.	N/A.

FR 5 — Restricted Data Flow (RDF)

Req (3-3 / 4-2)	Requirement	Where & how to verify (native steps + Ext. tools)	Pass criteria / evidence	Notes / lifecycle
SR/CR 5.1	Network segmentation	**Primary control.** The module **bridges the ControlLogix backplane to a Modbus TCP network**, and Modbus TCP + config share one Ethernet port - place it on a segmented OT cell behind a firewall, allow **502 only to/from authorized clients**, block config (PDS/web) from operational reach, and cross only required flows; no flat convergence with IT. **Ext. tools:** `nmap -p 502,80` per zone; firewall/allowed-client + config-path review.	Module on a segmented cell; 502 restricted; config path controlled; only required flows cross.	
SR 5.2	Zone boundary protection	Confirm zone-boundary protection — only required flows cross the boundary (system SR / network device). **Steps:** for this product this is assessed with the same evidence as **SR/CR 5.1** above (shared steps); confirm this requirement individually and adjudicate it separately.	Zone boundary protection verified individually.	
SR 5.3	General-purpose person-to-person comms restrictions	**N/A.** Single-purpose comms module. **Ext. tools:** Confirm no such services.	N/A - justification recorded.	N/A
SR 5.4	Application partitioning	Confirm the module cleanly partitions the ControlLogix (backplane) from the Modbus TCP network, and that config/diagnostics are separated (by firewall rule) from the operational Modbus TCP on the shared port. **Ext. tools:** Firewall rule review (502 vs config); backplane/Modbus partition.	Backplane/Modbus partitioned; config vs operational separated by rule.	

FR 6 — Timely Response to Events (TRE)

Req (3-3 / 4-2)	Requirement	Where & how to verify (native steps + Ext. tools)	Pass criteria / evidence	Notes / lifecycle
SR/CR 6.1	Audit log accessibility	No native module log - monitor off-module: the **ControlLogix processor + AssetCentre** for config/AOI changes, the module **LED/web diagnostics** + AOI status for health, and a Modbus-TCP-aware **OT IDS** on the cell (502 connection/write events); forward to the SIEM. **Ext. tools:** AssetCentre; module diagnostics/AOI status; Modbus/OT IDS + SIEM.	Module/Modbus TCP events monitored off-module in near-real-time.	No native log - Compensating.
SR/CR 6.2	Continuous monitoring	Confirm continuous monitoring — events forwarded to and monitored by a SIEM/OT-IDS in near-real-time. **Steps:** for this product this is assessed with the same evidence as **SR/CR 6.1** above (shared steps); confirm this requirement individually and adjudicate it separately.	Continuous monitoring verified individually.	

FR 7 — Resource Availability (RA)

Req (3-3 / 4-2)	Requirement	Where & how to verify (native steps + Ext. tools)	Pass criteria / evidence	Notes / lifecycle
SR/CR 7.1	Denial-of-service protection	Confirm resilience: a Modbus TCP flood or connection exhaustion (server = 10 connections) must not stop the backplane exchange or drive unsafe process action - verify the comm-fail behavior is safe; firewall/rate-limit 502; restrict clients. **Ext. tools:** ISOLATED bench: `hping3 --flood -p 502` / connection storm; confirm safe comm-fail + backplane continuity.	Backplane exchange continues (or fails safe) under Modbus-TCP DoS; 502 rate-limited/restricted.	

Req (3-3 / 4-2)	Requirement	Where & how to verify (native steps + Ext. tools)	Pass criteria / evidence	Notes / lifecycle
SR/CR 7.2	Resource management	Confirm bounded resources: the 10 server connections, client command limits, and the register map so one service cannot starve the exchange. **Ext. tools:** PCB (connection/command/register config); bench load test.	Resources bounded; no exhaustion-based loss of the exchange.	
SR/CR 7.3	Control system backup	Confirm the PCB **.ppf** + the Studio 5000 project (AOI/tags) + firmware record are backed up (versioned, offline) and a restore is tested on a spare module for a known-good rebuild. **Ext. tools:** Backup repository; `certutil -hashfile .ppf SHA256`; restore test (spare module).	Backups current and recovery validated.	
SR/CR 7.4	Control system recovery & reconstitution	Confirm recovery/reconstitution — a restore to a known-good state is tested. **Steps:** for this product this is assessed with the same evidence as **SR/CR 7.3** above (shared steps); confirm this requirement individually and adjudicate it separately.	Control system recovery & reconstitution verified individually.	
SR/CR 7.5	Emergency Power	**N/A.** Emergency/backup power is provided at the facility/UPS level, not the device; confirm the device retains its security state across a power transfer.	N/A - justification recorded (not assessed as a control).	N/A
SR/CR 7.6	Network & security configuration settings	Harden: **restrict Modbus TCP 502 to authorized clients**, **disable the on-board web**, disable unused client/server functions, set a non-default IP, restrict writable registers, and patch firmware; the ControlLogix keyswitch/FactoryTalk govern the processor side. **Ext. tools:** PCB (clients/registers/web); `nmap -p 502,80` (502 to authorized only; web off).	Least functionality; web disabled; 502 client-restricted; writes scoped; patched.	
SR/CR 7.7	Least functionality	Confirm least functionality — unused services/ports/protocols are disabled. **Steps:** for this product this is assessed with the same evidence as **SR/CR 7.6** above (shared steps); confirm this requirement individually and adjudicate it separately.	Least functionality verified individually.	
SR/CR 7.8	Component inventory	Inventory the module (MVI56E-MNET / MNETXT + firmware), its slot/chassis, and the IP/Modbus config, so vulnerabilities track against ProSoft / CISA advisories. **Ext. tools:** PCB / module LED (model/firmware); Studio 5000 (slot); asset register; ProSoft advisory check.	Accurate module/firmware inventory maintained; advisories tracked.	

External Test Tools — how to use

Tool / method	Verifies	How to use — steps / example	Constraints
ProSoft Configuration Builder (PCB)	Native config, client/register scope & diagnostics (most FRs)	Review the module config (IP, Modbus TCP client/server, allowed clients, register map), diagnostics, and firmware; back up the .ppf. Connect over the Ethernet port or via CIPconnect. Disable the on-board web.	Read/authorized project copy.
Studio 5000 AOI + FactoryTalk Security/AssetCentre	Processor-side authz & audit (1.1, 2.1, 2.8, 3.4)	The MVI56E-MNET is owned by the ControlLogix processor - review the AOI/controller tags, FactoryTalk Security roles, keyswitch, and AssetCentre change records.	Read-only; authorized.
modpoll / mbpoll (Modbus TCP)	Modbus TCP read-back & write-scope (1.1, 1.2, 2.1, 3.3)	`modpoll -m tcp -a` ; read status; confirm which registers are writable + that unauthorized clients are blocked (authorized/isolated).	Read-only on production; writes isolated.
nmap	Port exposure, 502 client-restriction & segmentation (1.13, 3.1, 5.x, 7.6)	`nmap -Pn -p 502,80` ; scan per zone; confirm 502 is reachable only from authorized clients and the web is off.	Read-only; authorized.
PDS / on-board web / curl	Config-surface check (1.1, 7.6)	ProSoft Discovery Service + `curl -i http:///`; confirm the web is disabled and config is reachable only via a controlled path/CIPconnect.	Read-only; authorized.
Wireshark / tcpdump	Comms integrity & containment (1.2, 3.1)	Filter `modbus`; confirm Modbus TCP stays in-zone to authorized clients; inspect writes; confirm no config exposure.	Passive; SPAN/tap only.
Modbus-TCP fuzzer / hping3 (isolated)	Input validation & DoS (3.5, 7.1)	`boofuzz` Modbus TCP to a BENCH module; `hping3 --flood -p 502` storm; watch LED + AOI status + backplane continuity.	ISOLATED bench; written authorization.

Tool / method	Verifies	How to use — steps / example	Constraints
Modbus-TCP OT IDS + ProSoft/CISA advisories	Monitoring & vuln tracking (2.8, 6.x, 7.8)	IDS on the cell for anomalous 502 connections/writes; track ProSoft / CISA advisories + required firmware.	Passive; coordinate with SOC.

Controlled - Project Use · Independent Assurance Services · ISA/IEC 62443-3-3 / 4-2 · Product-specific companion to the 62443 assessment tool; confirm exact menu paths / CLI against the deployed firmware and the project hardening baseline. ISOLATED tools run only in a representative/lab environment with written authorization.